Hy-Demosaicing: Hyperspectral Blind Reconstruction From Spectral Subsampling

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***Abstract*— This article proposes a smart hyperspectral sensing strategy, implemented in the spectral domain, conceived for spaceborne sensor systems, where physical space, storage resources, and communication bandwidth are extremely scarce and expensive. Smart sensing means faster and hardware-friendly imaging. Instead of acquiring all band samples in the spectral domain, we randomly select a few band samples per spatial pixel location. A periodic structure of spectral band selector array (SBSA) is designed so that we can learn a subspace basis from subsamples, which is essential to the underlying hyperspectral image (HSI) recovery algorithm. This spectral subsampling sensing strategy yields a demosaicing problem. We propose a blind hyperspectral reconstruction technique termed hyperspectral demosaicing (Hy-demosaicing) exploiting spectral low-rankness and spatial correlation of HSIs. It is blind in the sense that the signal subspace is learned from measured spectral subsamples. The subspace basis is data-adaptive and provides a more compact representation than other non-adaptive representations. This adaptiveness leads to improved image recovery as illustrated in experiments with real data.**

***Index Terms*— Demosaicing, plug-and-play, spectral imaging, subspace representation.**

I. Introduction

IN HYPERSPECTRAL imaging, the sensors measure the electromagnetic energy across an instantaneous field of view in hundreds (even thousands) of spectral channels. The very high spectral resolution of hyperspectral image (HSI) enables a precise identification of the sensed materials via spectroscopic analysis, facilitating countless applications [1].

However, the very high spectral resolution of HSIs also implies large volume of measurements, leads to a heavy burden on data acquisition, processing, transmission, and storage [2]. For example, the advanced hyperspectral imager (AHSI) aboard China's GaoFen-5 satellite [3] generates data at a rate of 2.0 Gbit/s. The downlink of GaoFen-5 satellite works at a rate of 900 Mbit/s. This example illustrates a challenge we are facing in spaceborne systems for Earth observation, due to the extremely large volumes of data collected by the imaging spectrometers onboard satellites, and the low bandwidth of the connections between the satellites and the ground stations.

The troubles caused by the huge volumes of data in HSIs have fostered the development of efficient and compact ways to deal with the involved information. This problem may be solved by the compressive sensing technology, since HSIs are inherently sparse and compressible due to their very high spectral and spatial correlation. For compressive hyperspectral sensing, data compression is implemented simultaneously with the acquisition, by computing a number of projections, termed measurements, of the original data onto a set of predesigned vectors. Assuming that the original data admits a sparse representation on a given basis or frame, it can be recovered from the projections by solving a suitable optimization problem [4]–[7].

For remotely hyperspectral compressive sensing systems, we refer to spectral compressive acquisition (SpeCA) [7], which computes random projections with the acquired spectral vectors, and hyperspectral-coded aperture (HYCA), which uses different random compression matrices (with Gaussian i.i.d. entries) for each pixel. These two measurement strategies use random compression matrices (with Gaussian/Bernoulli i.i.d. entries), i.e., projection patterns involving the complete spectra and thus requiring complex hardware systems. To avoid the burden on hardware design, we propose a spectral subsampling measurement strategy, called *Hy-demosaicing*, which is hardware-friendly. With our sensing strategy, it leads to higher ill-conditioned inverse problems, which require advanced solutions using computational methods.

We remark that Hy-demosaicing cannot be taken as a compressive sensing technique, but rather as a form of spectral bands subsampling. Conventional compressive sensing techniques transfer the data burden to the acquisition step in the sense of more complex hardware and longer acquisition times (such as single-pixel camera [8]) and to the signal recovery step [9]. For our specific application, remote-sensing hyperspectral imagers are placed on satellite/airborne platforms, meaning that we cannot afford to deploy heavy acquisition

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hardware and procedures. With these considerations, this article discusses a very fast hyperspectral sensing framework.

A complete sensing system contains not only a hardware system to acquire compressive or incomplete measurements, but also an algorithm to recover an image from measurements by solving a suitable optimization problem, which approximates the image sparsely on a basis or frame [10], [11]. For example, pseudo-inverse iterative hard thresholding (PIHT) reconstruction method [12] makes use of the sparsity of spectral vectors in a discrete cosine transform (DCT), and the sparsity of every image band in a lifting-based 2-D discrete wavelets transform (DWT). Kronecker compressed sensing (KCS) reconstruction method [13] exploits intrachannel spatial redundancy and interchannel correlation of multispectral images in the Fourier domain.

Apart from sparse representations, low-rank representation of HSIs has been successfully utilized for image recovery. For example, high accuracy low rank tensor completion (HaLRTC) [14], tensor nuclear norm (TNN) [15], and weighted sum of the tensor nuclear norm (WSTNN) [16] address tensor completion problem (i.e., filling missing elements of partially observed tensors) using low-rank tensor approximation. The fast HSI denoising method (FastHyDe) [17], nonlocal meets global (NG-meet) [18], and global local factorization (GLF) [19], and the adaptive HSI denoising method (AdHyde) [20] recover a noise-free HSI from its noisy observations by exploiting subspace representation of spectral/pixel vectors. The subspace representation can be interpreted as an explicit low-rank representation, but a key issue is to estimate the spectral subspace from observations. It has been studied in Section II-B1 of [21] that a spectral orthogonal subspace can be estimated easily by computing singular value decomposition (SVD) of Gaussian-noisy observations. However, the challenge to use subspace representation of spectral vectors in image recovery problem is that observations are compressed or incomplete. Usually, a spectral subspace cannot be directly learned from incomplete observations or compressed observations, those are linear combinations of original pixel vectors.

This work is motivated by the successful applications of subspace representation of spectral/pixel vectors in solving hyperspectral inverse problems [17], [22]–[27]. We develop a smart sensing strategy, which enable us to learn a spectral subspace directly and easily from observations.

A. Motivation

Our hyperspectral sensing strategy is inspired by color filter array (CFA) used in most modern digital red, green, blue (RGB) cameras, which acquires images using a single sensor overlaid with a CFA. Bayer filter is the most commonly used CFA configuration. It has alternating red and green filters for even rows and alternating green and blue filters for odd rows. The filter pattern is 50% green, 25% red, and 25% blue. The output of sensor is an array of pixel values, each indicating a raw intensity of one of the three filter colors. This is a subsampling strategy in both the spectral and spatial domain. Subsampling means to take a smaller portion of an original sample. For example, spectral subsampling of a pixel means to take a subset of its spectral measurements. These incomplete output of color samples can be exploited to reconstruct a full color image, which is called *demosaicing* [28]. The idea of using a CFA to sample RGB channels has also been extended to the multispectral domain. For example, in [29], a system uses a multispectral filter array (MSFA) to form from a square 4×4 basic pattern, and sample 16 wavelength bands. A binary tree-based MSFA is introduced in [30] to sample five wavelength bands. We refer to [31]–[33] for MSFAs with different patterns. It is noted that the above-mentioned patterns in MSFAs are fixed and not suitable for hyperspectral imaging with various number of wavelength bands.

B. Contributions

This article proposes a fast HSI sensing strategy, taking into consideration that HSIs are spectrally low-rank and spatially correlated. The sensing strategy is conceived to yield a demosaicing problem. The main contributions are given as follows.

1) A sensing strategy based on spectral subsampling (instead of random projections), which is compatible with the constraints imposed by the onboard acquisition hardware and acquisition geometry (such as pushbroom sensor). The sensing strategy is fast in the sense that it implements subsampling in both spectral and spatial domains, instead of doing random projections in conventional compressive sensing systems.
2) A periodic structure of spectral band selector array (SBSA) allows us to learn subspace basis from subsamples, which is essential to the recovery algorithm.

This work is a more fully developed extension based on our conference paper [34]. The identifiable benefit that this version offers to readers beyond the already published conference paper is given as follows.

1) Our spectral subsampling scheme is herein introduced and characterized in more detail. Several diagrams were added to explain how the subsampling scheme works. More experimental results were shown to illustrate the benefit of using our subsampling scheme.
2) Instead of using the GLF prior [34], the proposed image reconstruction method utilizes an image prior from a CNN-based denoising method, leading to faster image reconstruction.
3) Extensive experimental validation was carried out. Five representative or state-of-the-art reconstruction methods were compared to our method using one simulated HSI and three real HSIs.[1]

This article is organized as follows. Section II introduces our spectral subsampling strategy: Hy-mosaicing. Section III formulates our new denoising algorithm: Hy-demosaicing. Section IV presents the experimental results, including comparisons with the state-of-the-art methods. Section V concludes this article.

II. Spectral Subsampling Strategy: Hy-Mosaicing

Inspired by CFA used in RGB imaging, our hyperspectral sensing strategy acquires measurements by selecting sub-

[1] An MATLAB demo of this work is available at https://github.com/LinaZhuang for the sake of reproducibility.

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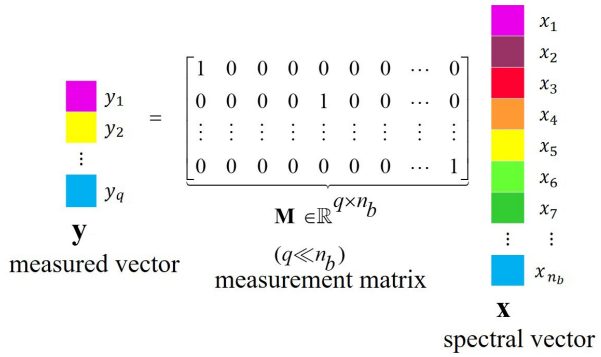


Fig. 1. Spectral subsampling acting on a pixel location, $\mathbf{x}$.

samples in spectral domain for every pixel location. The measurement matrix $\mathbf{M}$ is a binary matrix corresponding to a random subset of rows of the identity matrix, and acts on a spectral vector $\mathbf{x}$ of size n_b by acquiring q incomplete color subsamples, see Fig. 1. Spectral bands are selected uniformly at random. The benefit of using the random spectral band selection scheme, compared with predetermined schemes, is to enhance the representation ability of estimated subspace in the reconstruction. Experimental results will be reported in Section IV-E to illustrate the benefit of using the random scheme.

Here it comes to the question of how to design measurement matrix $\mathbf{M}$ for each pixel location. A straightforward strategy is random subsampling in both the spectral and spatial domains, meaning $\mathbf{M}$ can be different for each pixel location. In this article, a spatially structured SBSA, corresponding to a structured collection of measurement matrices, is proposed in our sensing strategy. The SBSA is spatially split the underlying HSI $\mathbf{X}$ into non-overlapping square windows of size $\sqrt{K}$-by-$\sqrt{K}$, where $\sqrt{K}$ is the size of the square window. For simplicity, we assume that the rows of $\mathbf{X}$ refer to spectral bands and the columns of $\mathbf{X}$ refer to spatial pixel locations with chronologically ordering. In this setting, the size of $\mathbf{X}$ is $n_b \times (n^2\ K)$, and the total number of square windows covering the whole image is n^2. In the proposed scheme, spatial pixel locations in all the square windows are replicated periodically. Thus there are K measurements in the SBSA and there are n^2 spectral vectors of size n_b involved in each measurement, i.e.,

$$\mathbf{X}_i = [\mathbf{x}_i, \ldots, \mathbf{x}_{\ell * n * K + j * K + i}, \ldots, \mathbf{x}_{(n-1)*n*K+(n-1)*K+i}]$$

where $1 \le i \le K$ and $\mathbf{x}_i \in \mathbb{R}^{n_b}$. In $\mathbf{X}_i$, the column indices ℓ and j are run from 0 to $(n-1)$, i.e., there are n^2 square windows. Thus, the size of $\mathbf{X}_i$ is n_b-by-n^2. It is clear that the underlying HSI can also be represented as follows:

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_K].$$

As an illustration of the proposed SBSA, we show an example of sensing strategy in Fig. 2. In the example, the size of HSI is 9-by-64, i.e., its spatial dimension is 8-by-8 and its spectral dimension is 9. The size of non-overlapping square window is 4-by-4, i.e., $\sqrt{K} = 4$ and the number of spectral bands to be randomly chosen is $q = 2$. It is noted that there are $n^2 = 4$ spectral vectors in each measurement, and there are 16 measurements $\mathbf{X}_1, \ldots, \mathbf{X}_{16}$ with two bands ($q = 2$) selected. For instance, $\mathbf{X}_1 = [\mathbf{x}_1, \mathbf{x}_5, \mathbf{x}_{33}, \mathbf{x}_{37}]$ because spatial pixel locations in the four windows are replicated periodically.

The spectral band sampling process of $\mathbf{X}_i$ can be formulated as follows:

$$\mathbf{Y}_i = \mathbf{M}_i \mathbf{X}_i, \quad i = 1, \ldots, K \tag{1}$$

where $\mathbf{M}_i \in \mathbb{R}^{q \times n_b} (q \ll n_b)$ represents color subsampling of the ith selector (sampling matrix) within the periodic cells, and $\mathbf{Y}_i \in \mathbb{R}^{q \times n^2}$ denotes measured subsamples. In Fig. 1, the ($K = 16$) measurement matrices $\mathbf{M}_i$ are set to obtain $\mathbf{Y}_i$. Let us consider additive measurement noise and vectorize measurements, the sensing process (1) can be formulated as

$$\mathbf{y} = \mathbf{M}\mathbf{x} + \gamma \tag{2}$$

where $\mathbf{x} := \mathrm{vec}(\mathbf{X}) = \mathrm{vec}([\mathbf{X}_1, \ldots, \mathbf{X}_K]) \in \mathbb{R}^{n_b n^2\ K}$, $\mathbf{y} := \mathrm{vec}([\mathbf{Y}_1, \ldots, \mathbf{Y}_K]) \in \mathbb{R}^{q n^2\ K}$, and $\gamma := \mathrm{vec}([\Gamma_1, \ldots, \Gamma_K]) \in \mathbb{R}^{q n^2\ K}$ denote HSI data, its measurements, and noise, respectively. The vec$(\cdot)$ operator stacks the columns of its matrix argument. Measurement matrix $\mathbf{M}$ is block diagonal

$$\mathbf{M} := \mathrm{bdiag}(\mathbf{M}_1, \ldots, \mathbf{M}_1, \mathbf{M}_2, \ldots, \mathbf{M}_2, \ldots, \mathbf{M}_K, \ldots, \mathbf{M}_K) \tag{3}$$

where bdiag$(\cdot)$ stands for a block diagonal matrix. Matrices $\mathbf{M}_i$ acts on each spectral vector by acquiring q incomplete color samples, implying a spectral subsampling with a measurement rate of q/n_b.

A possible hardware solution to implement the proposed sensing strategy on a pushbroom satellite sensor is given here. The pushbroom scheme is a widely used scanning strategy in HSIs for remote sensing. Fig. 3(a) conceptually depicts an imaging process of a pushbroom hyperspectral imaging spectrometer, which is a line-scan system and an entire line of the HSI is acquired at once, with the motion of the satellite platform providing the along-track scanning [35]. A pushbroom architecture for the Hy-mosaicing acquisition of HSI is depicted in Fig. 3(b), where a spectral-splitting device (e.g., a prism) spreads a single line from a slit aperture across the digital micromirror device (DMD)[2] array. The subsampling in the spectral domain is performed with the DMD array, where each element determines whether the arriving light is acquired or not by changing the micromirror state ("on" and "off"). The reflections of light from DMD array are gathered in a linear array of photosensors through a cylindrical lens. To implement color selection (instead of projection), we set a special measurement pattern on the DMD array, i.e., for each row, there is only one micromirror having "on" state so that only that color is acquired finally in the linear array of photosensors. This process is repeated q times with different DMD patterns so that each pixel has q color subsamples.

III. Image Reconstruction: Hy-Demosaicing

The reconstruction of the HSI from the subsampled measurements (2) is a hyperspectral demosaicing problem.

[2]Introduction to DMD: https://www.ti.com/lit/an/dlpa008b/dlpa008b.pdf

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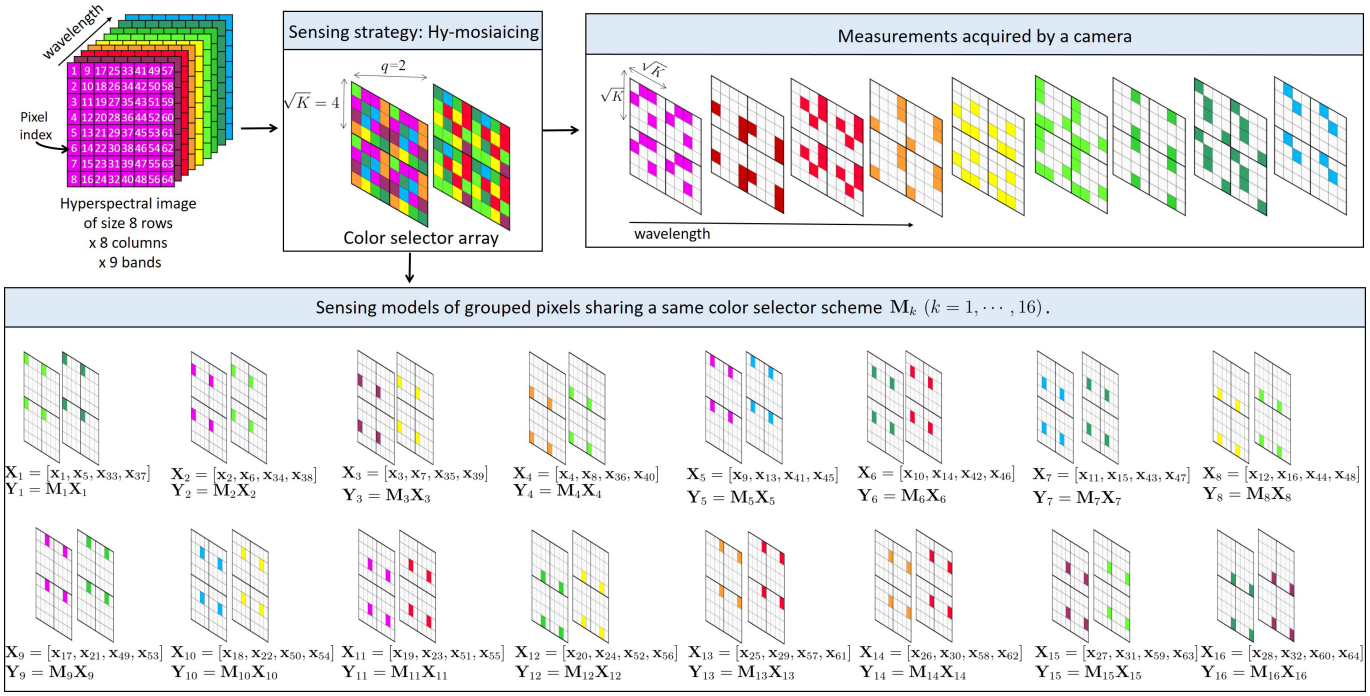


Fig. 2. Proposed sensing strategy: Hy-mosaicing.

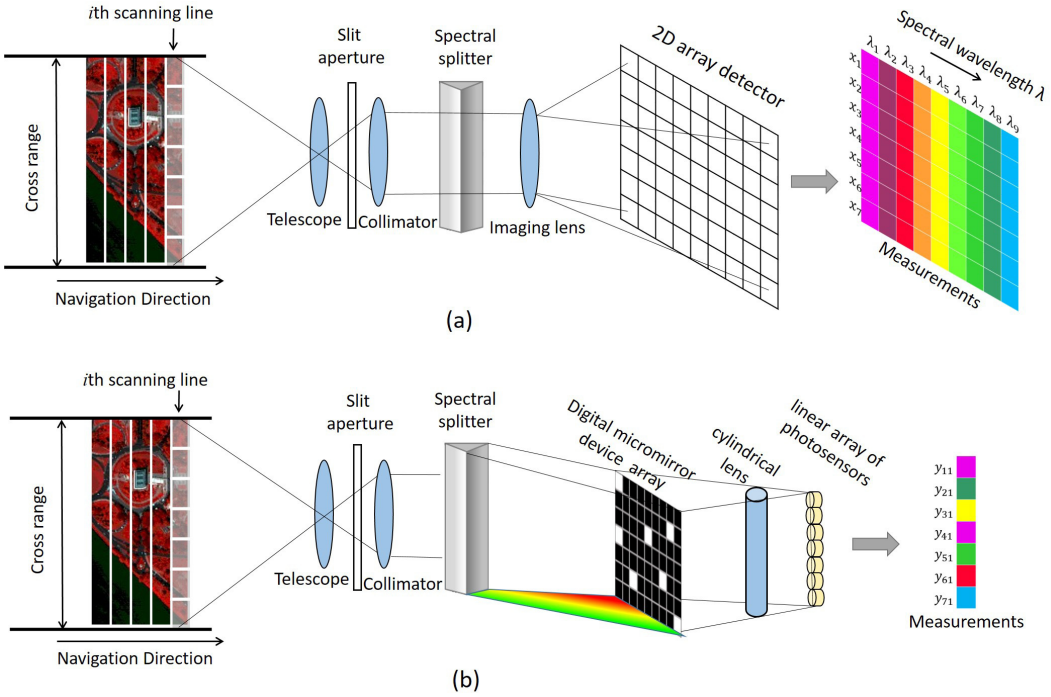


Fig. 3. (a) Imaging process of a pushbroom hyperspectral imaging spectrometer. (b) Possible solution for implementing the de-mosaicing sensing strategy in a pushbroom satellite sensor.

To solve this problem, we take advantage of two characteristics of HSIs: 1) the high correlation among spectral channels, which can be exploited using subspace representation of the spectral vectors and 2) the strong correlation existing in the spatial domain, which is exploited by promoting spatial correlation of subspace representation coefficients.

A. Subspace Representation

Due to the extremely high spectral correlation, spectral vectors in the HSI admit a subspace representation [1], [19], i.e.,

$$\mathbf{x} = \text{vec}(\mathbf{EZ}) = \mathbf{I} \otimes \mathbf{E}\text{vec}(\mathbf{Z}) = \mathbf{I} \otimes \mathbf{Ez} \tag{4}$$

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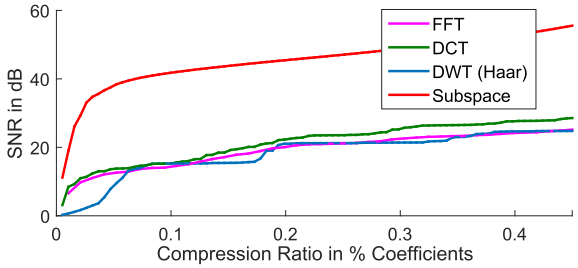


Fig. 4. Reconstruction performance with assumptions that 1-D spectral vectors are sparse in different transform domains. SNR: $= 10\log_{10}(\mathbb{E}\|\mathbf{X}\|_F^2/\mathbb{E}\|\mathbf{X}-\widehat{\mathbf{X}}\|_F^2)$, where $\mathbf{X}$ is image matrix of Washington DC Mall HSI and $\widehat{\mathbf{X}}$ is its reconstruction.

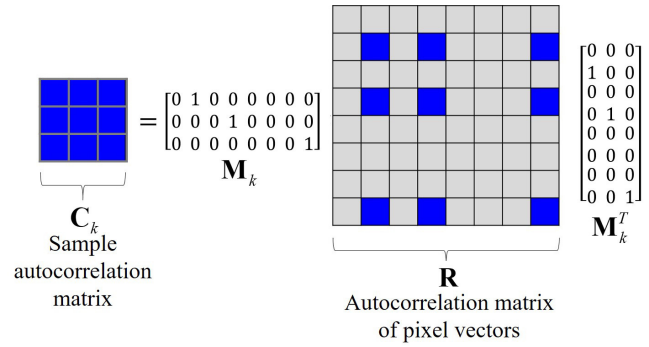


Fig. 5. Relation between autocorrelation matrix of pixel vectors and the autocorrelation matrix of sampled vectors.

where $\mathbf{I} \in \mathbb{R}^{(n^2\ K)\times(n^2\ K)}$ is an identity matrix, columns of $\mathbf{E} \in \mathbb{R}^{n_b\times p}(p < q)$ span an orthogonal subspace, and $\mathbf{z}$ is the vectorized representation of the coefficients.

The motivation of using subspace representation instead of other sparse representations for spectral vectors is illustrated in Fig. 4. We take the real HSI, Washington DC Mall image, as an example. Because of the high correlation among the spectral channels, the spectral vectors can be sparsely represented in some transform domains, such as the fast Fourier transform (FFT), DCT, and DWT. Fig. 4 shows that using subspace representation achieves better reconstruction performance than other bases. The main reason is that the subspace basis is data-adaptive and thus yields a more compact representation of the HSI, whereas the compared bases are non-adaptive to data. The advantage of subspace representation has also been studied for HSI denoising [17], [21]. In demosaicing problem, the challenge of exploiting subspace representation lies in the estimation of the subspace from incomplete observations in the spectral domain.

B. Subspace Estimation

One of the contributions of this work is to estimate the spectral subspace from incomplete samples in the spectral domain. A matrix spanning the same subspace as $\mathbf{E}$ can be learned from autocorrelation matrix of the spectral vectors, denoted by $\mathbf{R} \in \mathbb{R}^{n_b\times n_b}$.

The autocorrelation matrix $\mathbf{R}$ of spectral vectors cannot be obtained directly from the measurements. However, thanks to the spatially structure designed in SBSA [see (3)], a relation between $\mathbf{R}$ and the sample autocorrelation matrix, denoted as $\mathbf{C}_k \in \mathbb{R}^{q\times q}$ for $k = 1, \ldots, K$, can be derived

$$\begin{aligned}\mathbf{C}_k &= \mathbb{E}\left[\mathbf{Y}_k\mathbf{Y}_k^T\right]\\ &= \mathbb{E}\left[\mathbf{M}_k\mathbf{X}_k\mathbf{X}_k^T\mathbf{M}_k^T\right]\\ &= \mathbf{M}_k\mathbb{E}\left[\mathbf{X}_k\mathbf{X}_k^T\right]\mathbf{M}_k^T\\ &= \mathbf{M}_k\mathbf{R}\mathbf{M}_k^T, \quad (k = 1, \ldots, K)\end{aligned} \tag{5}$$

where $\mathbb{E}$ denotes an averaging operator, and columns of $\mathbf{Y}_k$ are measurements of pixels sharing the same measurement matrix $\mathbf{M}_k$. This relation is depicted in Fig. 5.

Given the matrices $\mathbf{C}_k$, for $k = 1, \ldots, K$, we approximate the autocorrelation matrix of spectral vectors by computing

$$\begin{aligned}\widehat{\mathbf{R}} &= \arg\min_{\mathbf{R}} \sum_{k=1}^{K} \left\|\mathbf{C}_k - \mathbf{M}_k\mathbf{R}\mathbf{M}_k^T\right\|_F^2\\ &= \left(\sum_{k=1}^{K}\mathbf{M}_k^T\mathbf{C}_k\mathbf{M}_k\right) \oslash \left(\sum_{k=1}^{K}\mathbf{a}_k\mathbf{a}_k^T\right)\end{aligned} \tag{6}$$

where $\mathbf{a}_k = \mathrm{diag}(\mathbf{M}_k^T\mathbf{M}_k)$, and $\oslash$ stands for element-wise division (see [36] for more details about above solution).

Claim 1: Relying on the *law of large numbers*, optimization (6) yields asymptotically the exact autocorrelation matrix if two conditions are satisfied: 1) the number of measurements corresponding to each measurement matrix $\mathbf{M}_k$ tends to infinity and 2) $\sum_{k=1}^{K}(\mathbf{a}_k\mathbf{a}_k^T) > 0$, ($\mathbf{a}_k = \mathrm{diag}(\mathbf{M}_k^T\mathbf{M}_k)$, and $>$ denotes element-wise greater than), meaning that every element of the autocorrelation matrix $\widehat{\mathbf{R}}$ is sampled at least once in each color cell.

The two above-mentioned conditions provide insights into the design of the mosaicing scheme, which is discussed in Section IV-D.

Once the autocorrelation matrix is estimated from measurements, we set

$$\widehat{\mathbf{E}} = \mathbf{U}(:, 1:p) \tag{7}$$

where $\mathbf{U}(:, i)$ is the eigenvectors of $\widehat{\mathbf{R}}$ corresponding the ith largest eigenvalues.

C. Hy-Demosaicing

Given the estimate of spectral subspace, the HSI can be reconstructed by solving the following optimization problem:

$$\widehat{\mathbf{z}} = \arg\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{M}(\mathbf{I}\otimes\widehat{\mathbf{E}})\mathbf{z} - \mathbf{y})\|^2 + \lambda\phi(\mathbf{z}) \tag{8}$$

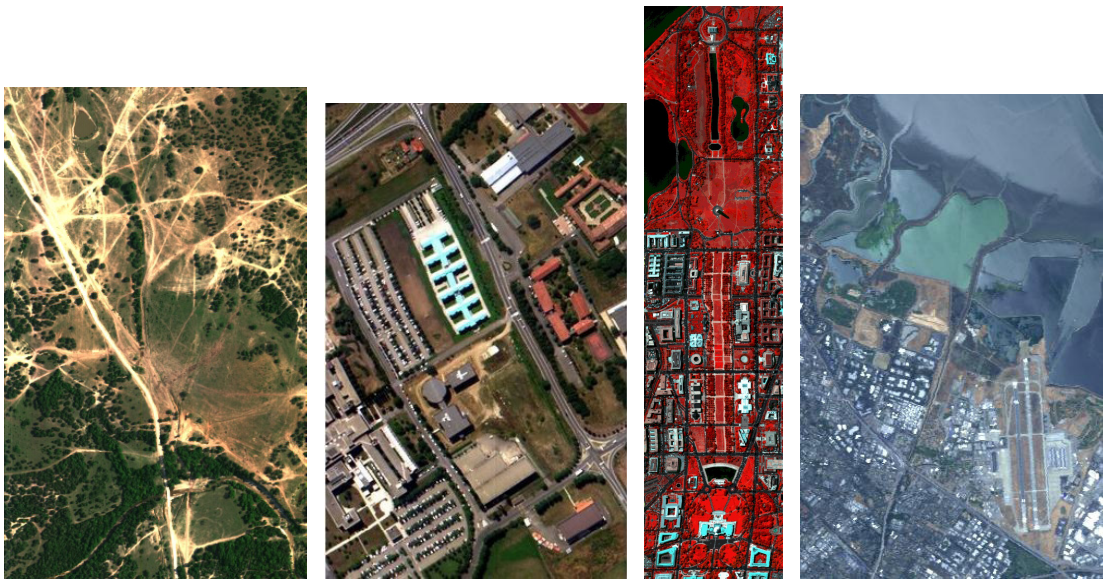
where the first term on the right-hand side represents the data fidelity and accounts for the Gaussian noise, and $\phi(\cdot)$ is a regularizer expressing prior information tailored to spatially correlated representation coefficients.

Solvers for optimization problem (8) will be iterative due to a non-diagonal operator involved in (8). To sidestep iterations and speed up the algorithm, (8) is solved approximately in two steps.

In first step, we estimate the missing components at each pixel by solving a simple least square problem

$$\begin{aligned}\tilde{\mathbf{z}}_i &= \arg\min_{\mathbf{z}_i} \|\mathbf{y}_i - \mathbf{M}_i\mathbf{E}\mathbf{z}_i\|_2\\ &= \left(\mathbf{E}^T\mathbf{M}_i^T\mathbf{M}_i\mathbf{E}\right)^{-1}\mathbf{E}^T\mathbf{M}_i^T\mathbf{y}_i\end{aligned} \tag{9}$$

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(a) (b) (c) (d)

Fig. 6. HSIs used in the experiments. (a) HYDICE data terrain image. (b) ROSIS data Pavia University image. (c) HYDICE data Washington DC Mall image. (d) Sub-image of AVIRIS data Moffett field.

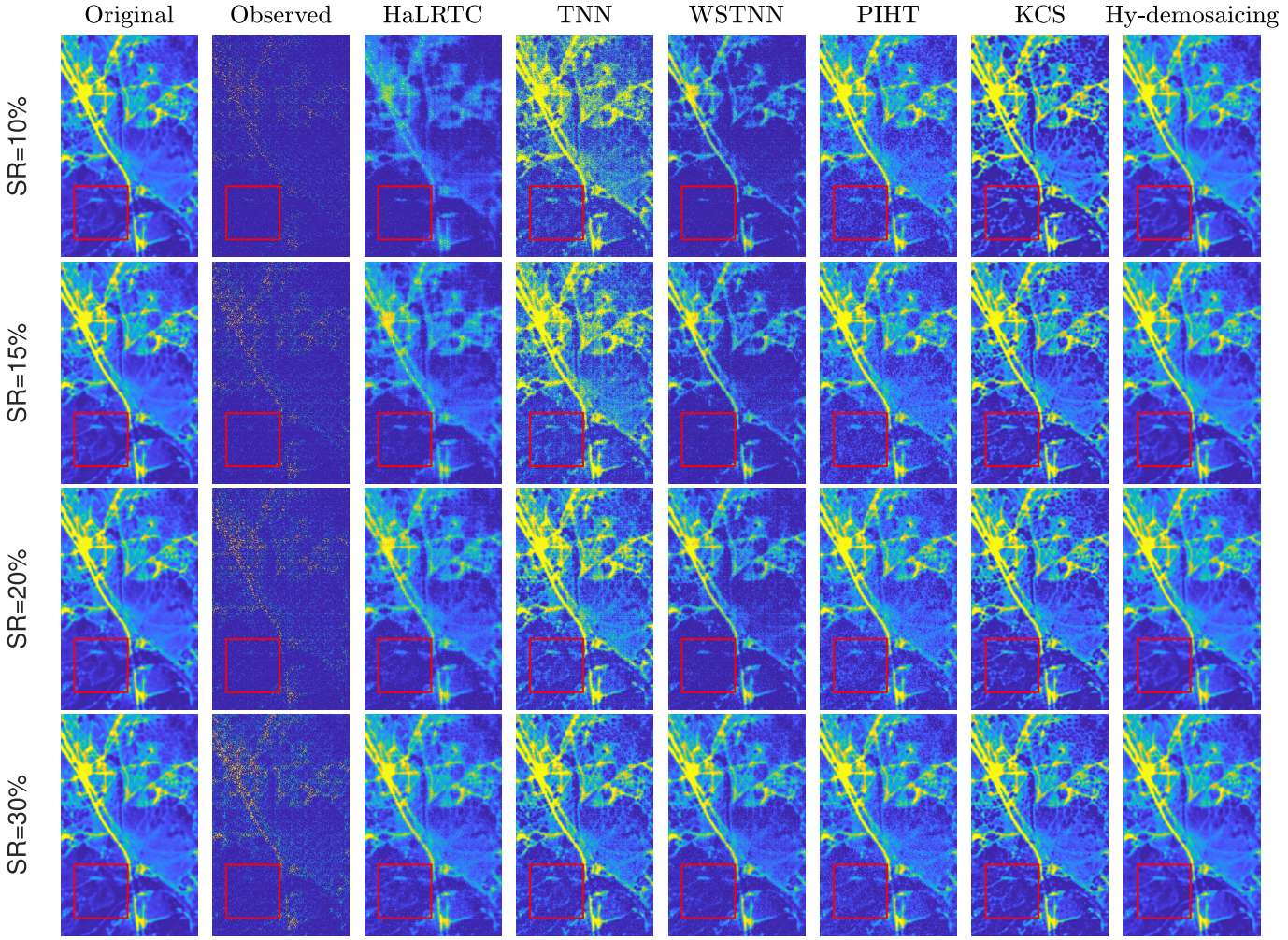


Fig. 7. Original band 5 of terrain data and its reconstructions from measurements. Measurements are obtained by a Hy-mosaicing strategy with $q = 50$, $\sqrt{K} = 30$, and $\sigma = 0.04$.

where the matrix $\mathbf{E}^T\mathbf{M}_i^T\mathbf{M}_i\mathbf{E}$ is non-singular when $q \geq p$. Given $\tilde{\mathbf{z}}_i$, we can estimate the unobserved component at each pixel by computing

$$\tilde{\mathbf{y}}_i = \mathbf{E}\tilde{\mathbf{z}}_i. \tag{10}$$

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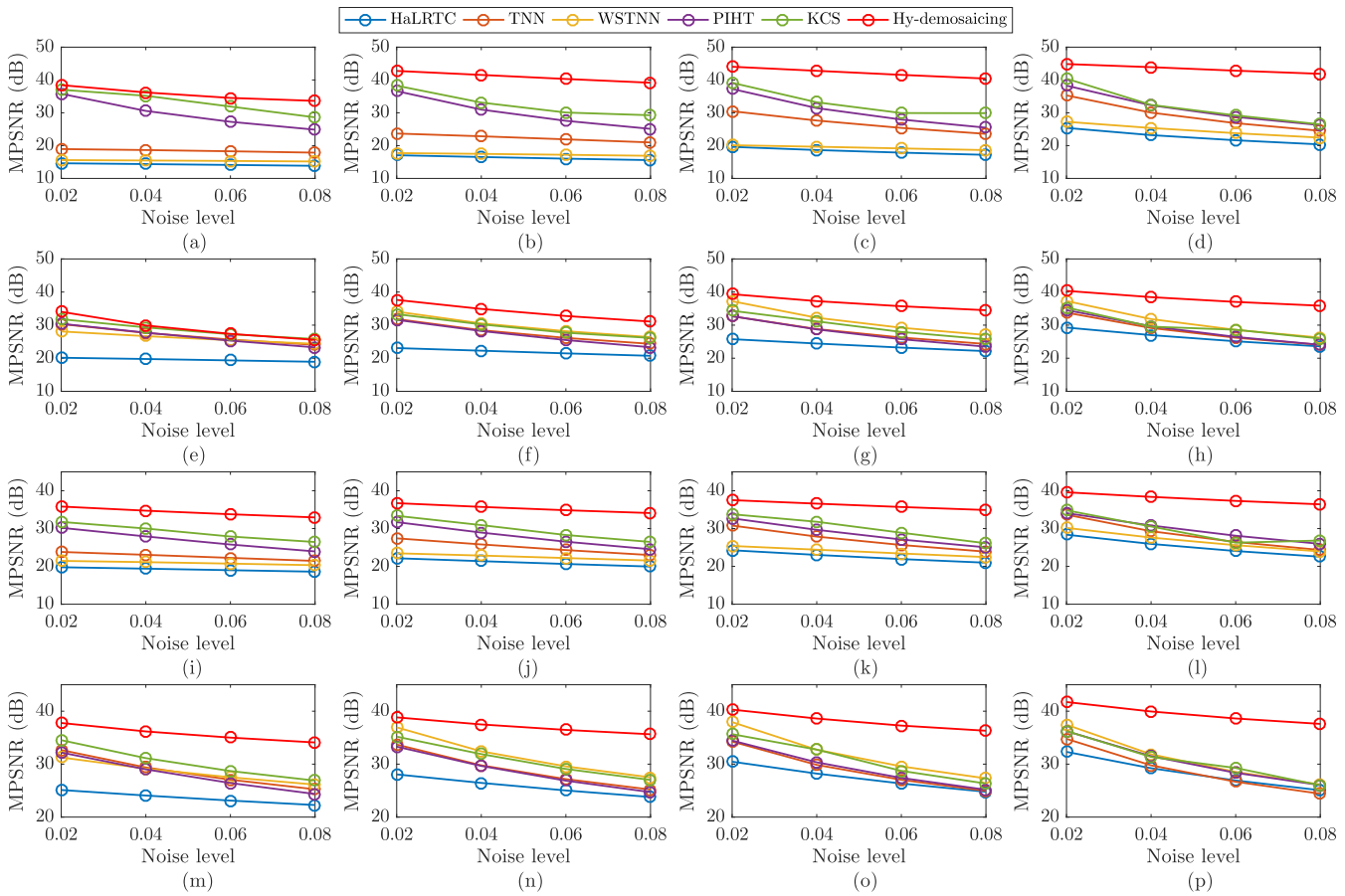


Fig. 8. MPSNRs of reconstructed images. Terrain with (a) SR=10%, (b) SR=15%, (c) SR=20%, and (d) SR=30%, Pavia University with (e) SR=10%, (f) SR=15%, (g) SR=20%, and (h) SR=30%, Washington DC Mall with (i) SR=10%, (j) SR=15%, (k) SR=20%, and (l) SR=30%, Moffett field with (m) SR=10%, (n) SR=15%, (o) SR=20%, and (p) SR=30%.

Since the estimates of the missing components are noisy, in second step, we perform a denoising step on $\widetilde{\mathbf{Y}} = [\tilde{\mathbf{y}}_1, \ldots, \tilde{\mathbf{y}}_{(n^2\ K)}]$

$$\widehat{\mathbf{Z}} = \arg\min_{\mathbf{Z}} \frac{1}{2}\|\widetilde{\mathbf{Y}} - \widehat{\mathbf{E}}\mathbf{Z}\|_F + \lambda\phi(\mathbf{Z}) \tag{11}$$

$$= \arg\min_{\mathbf{Z}} \frac{1}{2}\|\widehat{\mathbf{E}}^T\widetilde{\mathbf{Y}} - \mathbf{Z}\|_F + \lambda\phi(\mathbf{Z}) \tag{12}$$

which is a proximity operator of ϕ applied to $\mathbf{Z}' = \widehat{\mathbf{E}}^T\widetilde{\mathbf{Y}}$. A proof of the equivalence of (11) and (12) can be found in Appendix A of [19]. Considering that orthogonal projection is a decorrelation transformation and rows of $\mathbf{Z}$ tend to be decorrelated, we decouple $\phi(\cdot)$ with respect to the rows, i.e.,

$$\widehat{\mathbf{Z}}(i,:) = \arg\min_{\mathbf{Z}(i,:)} \frac{1}{2}\|\mathbf{Z}'(i,:) - \mathbf{Z}(i,:)\|_F + \lambda\phi\big(\mathbf{Z}(i,:)\big) \quad i = 1, \ldots, p \tag{13}$$

where $\mathbf{Z}(i,:)$ represents ith row of $\mathbf{Z}$. To solve subproblem (13), we resort to the plug-and-play trick [21], [26], [37], whose main idea is to directly use an existing regularizer from a state-of-the-art denoiser, instead of investing effort in designing a new regularizer promoting spatially correlated eigen images. In this article, the prior of a CNN-based denoising network, FFDnet [38], is plugged in (13), leading to

$$\widehat{\mathbf{Z}}(i,:) \leftarrow \text{FFDNet}(\mathbf{Z}'(i,:)) \tag{14}$$

where the function FFDNet$(\cdot)$ outputs a denoised eigen-image. The FFDNet network contains a series of 3×3 convolution layers, which are composed of three types of operations: convolution, rectified linear units, and batch normalization [38]. Also, the input of FFDNet network requires the parameter of noise level so that the network can process images with different noise levels. We remark that other state-of-the-art single band denoisers [such as block-matching and 3-D filtering (BM3D) [39] and weighted nuclear norm minimization (WNNM) [40]] also can be adopted to estimate $\mathbf{Z}(i,:)$. The FFDNet is selected in this work due to its following advantages over others. Compared with other machine learning-based denoisers [39], [40], deep-learning-based FFDNet is much faster as long as it has been well trained. Compared with other deep-learning-based denoisers [41], [42], FFDNet is able to address images with various noise levels, meaning that we can input a new image without retraining.

The final estimate of the HSI is obtained by computing

$$\widehat{\mathbf{x}} = \mathbf{I} \otimes \widehat{\mathbf{E}}\hat{\mathbf{z}} \tag{15}$$

where $\hat{\mathbf{z}} = \text{vec}(\widehat{\mathbf{Z}})$.

IV. Experiments

We conduct experiments using a simulated HSI, called terrain, and four real HSIs, namely, the Pavia University

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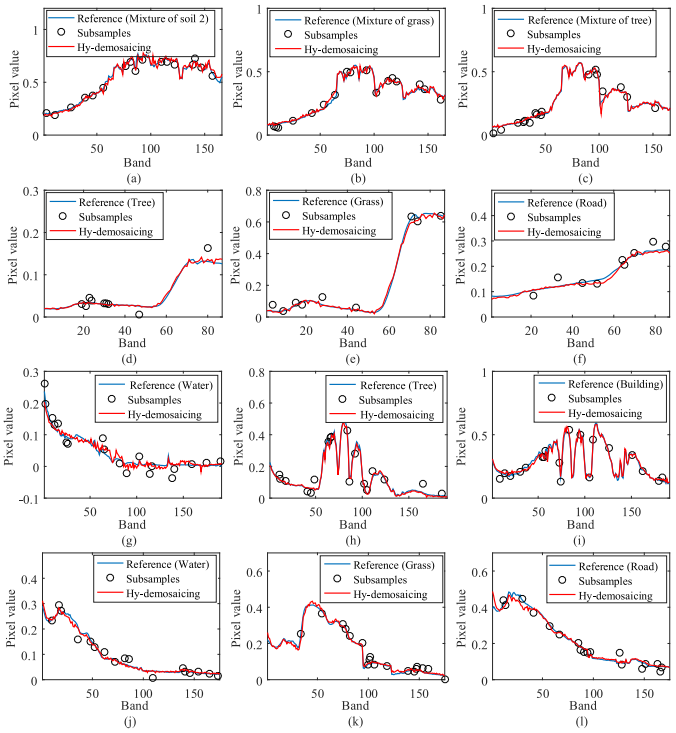


Fig. 9. Spectra of representative materials reconstructed by Hy-demosaicing method in (a)–(c) terrain data, (d)–(f) Pavia University data, (g)–(i) Washington DC Mall data, and (i)–(l) Moffett field data. Subsamples marked by black circles are measurements acquired by the proposed Hy-mosaicing sensing strategy (when SR = 10%, $\sqrt{K} = 30$, and $\sigma = 0.02$).

image[3] of size 610 rows × 340 columns × 87 bands, the Washington DC Mall image[4] of size 1208 rows × 307 columns, the AVIRIS Moffett field data[5] of size 500 rows × 307 columns × 166 bands, and HYDICE urban data[6] of size 307 rows × 307 columns × 162 bands. The four real HSIs are well-known public data with high-quality spectral and spatial resolution, shown in Figs. 6 and 15.

A. Test Images and Compared Image Reconstruction Methods

To evaluate the impact of image reconstruction error on the hyperspectral unmixing task, we simulated a clean semi-real HSI based on the public available terrain image of size 500 rows × 307 columns × 166 bands [Fig. 6(a)] following the generation steps in [25].[7] The terrain image is mainly composed of soil, tree, grass, and shadows. The number of endmembers is empirically set to five, as it was set in [25], [44], and [45]. Briefly, a clean terrain image is synthesized based on the linear mixing model, i.e., $\mathbf{X} = \mathbf{AM}$, where $\mathbf{M} \in \mathbb{R}^{n_b \times 5}$ and $\mathbf{A} \in \mathbb{R}^{5 \times (n^2\ K)}$ are endmembers and abundances, respectively, estimated from the original terrain image.

[3]http://www.ehu.eus/ccwintco/uploads/e/ee/PaviaU.mat

[4]https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

[5]https://aviris.jpl.nasa.gov/data/free_data.html

[6]https://rslab.ut.ac.ir/data

[7]An MATLAB demo describing image simulation can be found in https://github.com/LinaZhuang/NMF-QMV_demo

TABLE I

IMAGE RECONSTRUCTION PERFORMANCE OF TERRAIN DATA ACQUIRED BY HY-MOSAICING STRATEGY (WITH $\sqrt{K} = 30$ AND $\sigma = 0.02$)

SR	Indexes	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-demosaicing
10%	**MPSNR**	14.62	18.95	15.59	35.79	37.02	**38.43**
	MSSIM	0.25	0.34	0.24	0.91	0.96	**0.97**
	MFSIM	0.67	0.70	0.69	0.94	0.97	**0.98**
	Time	283	889	1988	577	450	**82**
15%	**MPSNR**	17.07	23.68	17.72	36.61	38.26	**42.76**
	MSSIM	0.33	0.54	0.32	0.92	0.95	**0.99**
	MFSIM	0.70	0.80	0.70	0.94	0.97	**0.99**
	Time	284	918	2068	587	454	**81**
20%	**MPSNR**	19.63	30.48	20.12	37.27	39.14	**44.03**
	MSSIM	0.41	0.78	0.41	0.92	0.96	**0.99**
	MFSIM	0.74	0.91	0.74	0.95	0.98	**0.99**
	Time	302	952	2148	597	459	**84**
30%	**MPSNR**	25.39	35.31	27.26	38.27	40.40	**44.86**
	MSSIM	0.60	0.88	0.67	0.94	0.97	**0.99**
	MFSIM	0.83	0.95	0.86	0.96	0.98	**0.99**
	Time	309	583	2174	597	462	**86**

TABLE II

IMAGE RECONSTRUCTION PERFORMANCE OF PAVIA UNIVERSITY DATA ACQUIRED BY HY-MOSAICING STRATEGY (WITH $\sqrt{K} = 30$ AND $\sigma = 0.02$)

SR	Indexes	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-demosaicing
10%	**MPSNR**	20.14	30.33	28.10	30.42	31.79	**34.03**
	MSSIM	0.45	0.79	0.77	0.81	0.86	**0.91**
	MFSIM	0.69	0.90	0.93	0.89	0.92	**0.95**
	Time	122	267	676	267	193	**59**
15%	**MPSNR**	23.07	31.84	34.08	31.59	33.31	**37.63**
	MSSIM	0.58	0.83	0.91	0.84	0.90	**0.96**
	MFSIM	0.76	0.92	0.96	0.91	0.94	**0.98**
	Time	128	285	709	239	191	**58**
20%	**MPSNR**	25.76	32.69	37.23	32.70	34.34	**39.35**
	MSSIM	0.68	0.85	0.94	0.86	0.91	**0.97**
	MFSIM	0.82	0.93	0.98	0.93	0.95	**0.98**
	Time	111	240	596	237	192	**59**
30%	**MPSNR**	29.27	33.78	37.28	34.55	35.33	**40.35**
	MSSIM	0.80	0.87	0.93	0.89	0.91	**0.98**
	MFSIM	0.88	0.94	0.97	0.95	0.96	**0.99**
	Time	71	248	292	249	196	**59**

TABLE III

IMAGE RECONSTRUCTION PERFORMANCE OF WASHINGTON DC MALL DATA ACQUIRED BY HY-MOSAICING STRATEGY (WITH $\sqrt{K} = 30$ AND $\sigma = 0.02$)

SR	Indexes	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-demosaicing
10%	**MPSNR**	19.77	23.81	21.45	30.20	31.76	**35.82**
	MSSIM	0.38	0.56	0.40	0.80	0.85	**0.95**
	MFSIM	0.71	0.83	0.87	0.89	0.92	**0.97**
	Time	566	1392	6224	2006	845	**109**
15%	**MPSNR**	22.15	27.42	23.46	31.70	33.37	**36.72**
	MSSIM	0.51	0.71	0.54	0.84	0.89	**0.96**
	MFSIM	0.78	0.88	0.88	0.91	0.94	**0.97**
	Time	582	1394	6640	1916	770	**110**
20%	**MPSNR**	24.26	30.79	25.40	32.71	33.78	**37.52**
	MSSIM	0.60	0.82	0.64	0.87	0.89	**0.96**
	MFSIM	0.83	0.92	0.89	0.93	0.95	**0.98**
	Time	598	1370	6759	2853	1384	**112**
30%	**MPSNR**	28.38	33.67	30.18	34.08	34.83	**39.59**
	MSSIM	0.76	0.88	0.81	0.89	0.91	**0.97**
	MFSIM	0.89	0.94	0.93	0.94	0.96	**0.99**
	Time	589	831	6922	2722	1365	**116**

The proposed sensing strategy was evaluated on the four test images. Additive Gaussian i.i.d. noise is added to measurements with noise level (i.e., standard deviation of noise) {0.02, 0.04, 0.06, and 0.08}. To recover an image from incomplete measurements, we used reconstruction methods, including HaLRTC [14], TNN [15], WSTNN [16], PIHT [12], KCS [13],

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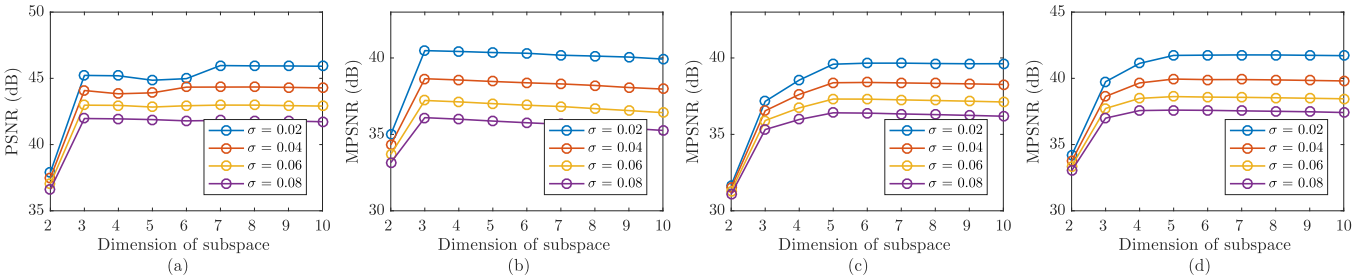


Fig. 10. MPSNR values of (a) terrain data, (b) Pavia University data, (c) Washington DC Mall data, and (d) Moffett field data recovered by the proposed Hy-demosaicing method as a function of p (the dimension of subspace). Measurements were acquired using the Hy-mosaicing strategy with $K = 30 \times 30$ and SR = 30%.

TABLE IV

IMAGE RECONSTRUCTION PERFORMANCE OF MOFFETT FIELD DATA ACQUIRED BY HY-MOSAICING STRATEGY (WITH $\sqrt{K} = 30$ AND $\sigma = 0.02$)

SR	Indexes	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-demosaicing
10%	**MPSNR**	25.12	32.66	31.26	32.19	34.52	**37.77**
	MSSIM	0.60	0.81	0.81	0.81	0.87	**0.95**
	MFSIM	0.75	0.90	0.93	0.88	0.92	**0.96**
	Time	140	348	1102	293	206	**55**
15%	**MPSNR**	28.06	33.69	36.95	33.30	35.19	**38.90**
	MSSIM	0.70	0.83	0.92	0.84	0.88	**0.96**
	MFSIM	0.80	0.92	0.96	0.90	0.94	**0.97**
	Time	141	321	1120	354	207	**61**
20%	**MPSNR**	30.46	34.26	37.94	34.46	35.66	**40.27**
	MSSIM	0.77	0.84	0.92	0.87	0.89	**0.97**
	MFSIM	0.84	0.93	0.97	0.92	0.94	**0.98**
	Time	134	308	645	317	212	**57**
30%	**MPSNR**	32.32	34.78	37.45	36.23	36.25	**41.75**
	MSSIM	0.82	0.85	0.91	0.90	0.89	**0.98**
	MFSIM	0.89	0.93	0.96	0.94	0.95	**0.98**
	Time	53	307	581	339	215	**57**

and the proposed Hy-demosaicing method.[8] To make a quantitative assessment, the peak signal-to-noise ratio (PSNR) index, the structural similarity (SSIM) index [45], and the feature similarity (FSIM) index [46] of each band were calculated. The corresponding mean PSNR (MPSNR), mean SSIM (MSSIM), and mean FSIM (MFSIM) are given in Tables I–IV for comparison.

B. HSI Reconstruction Performance

The image reconstruction performance will be discussed with various settings of the Hy-mosaicing sensing strategy, namely q, noise level, and K. First, we perform experiments with the sampling ratio (SR) {10%, 15%, 20%, 30%} by fixing $K = 30 \times 30$ and changing the values of q. The quality of four images recovered by different methods are measured in Tables I–IV, respectively. To save space, only one band in the terrain data and its reconstruction results are presented in Fig. 7. Basically, four HSIs can be recovered with higher quality when sampling ratio is higher. Compared with other image reconstruction methods, the proposed Hy-demosaicing method uniformly yields the best results in Tables I–IV. HaLRTC, TNN, and WSTNN are state-of-the-art low rank tensor completion methods using different kinds of rank definition for tensors. A comparison with HaLRTC, TNN, and WSTNN methods is to show that the Hy-demosaicing problem cannot be simply solved as a tensor completion problem. PIHT makes use of the sparsity of spectral vectors in a DCT, and the sparsity of every image band in a lifting-based 2-D DWT. KCS exploits intrachannel spatial redundancy and interchannel correlation of multispectral images in the Fourier domain. Hy-demosaicing yields uniformly the best performance in Tables I–IV in terms of MPSNR, MSSIM, and MFSIM, demonstrating data-adaptive basis learned from subsamples are useful to recover the HSI with limited number of measurements. Furthermore, by sidestepping the iteration, Hy-demosaicing method is much faster than other methods.

Apart from the sampling ratio, we also discuss the impact of noise level (i.e., standard deviation of Gaussian noise) on the image reconstruction. Fig. 8 provides MPSNRs of reconstructed images as a function of standard deviation σ of Gaussian noise. Four images and four kinds of sampling

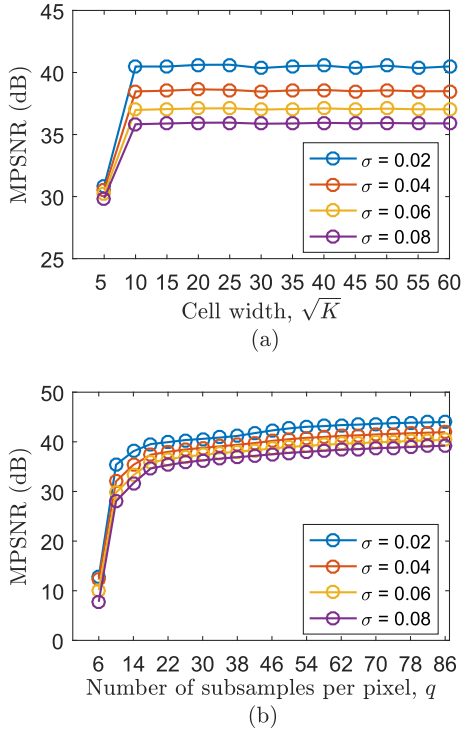


Fig. 11. MPSNR values of Pavia University data recovered by the proposed Hy-demosaicing method as a function of the width of square cells and number of subsamples per pixel.

[8]MATLAB code of Hy-demosaicing is available on the GitHub page https://github.com/LinaZhuang

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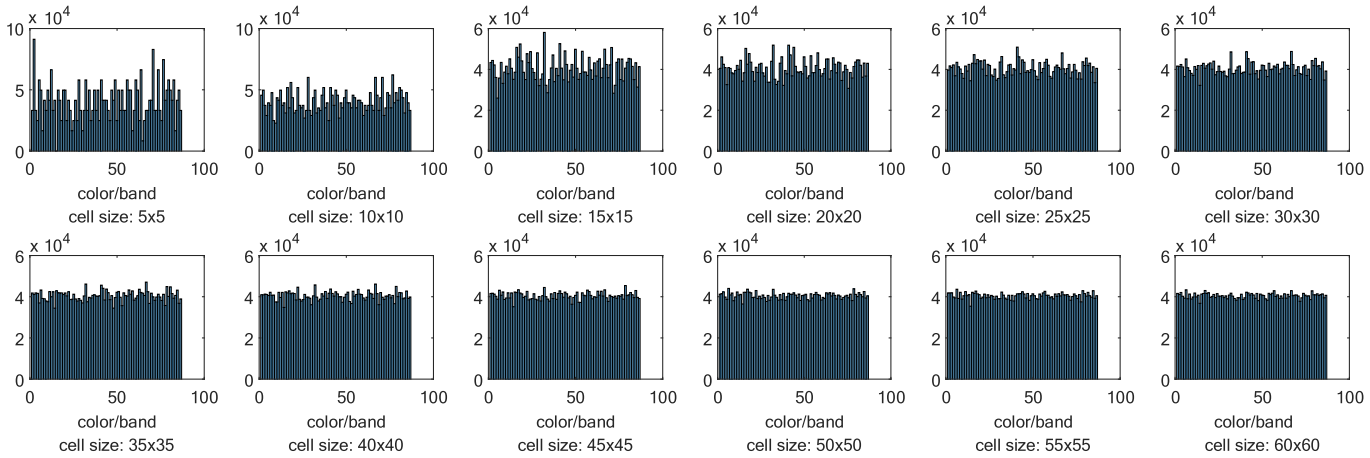


Fig. 12. Histograms of colors acquired by our subsampling scheme with different cell size and same q in Pavia University data.

TABLE V

PERFORMANCE COMPARISON OF TWO SUBSAMPLING SCHEMES IMPLEMENTED TO ACQUIRE PAVIA UNIVERSITY DATA WITH $\sqrt{K} = 30$, AND $q = 9$

	A pre-determined subsampling scheme (equally distributed)	Proposed subsampling scheme (random)
MPSNR of $\widehat{\mathbf{x}}$	-30 dB	38 dB
$\gamma = \frac{\|\widehat{\mathbf{E}}\widehat{\mathbf{E}}^T\mathbf{X}\|_F^2}{\|\mathbf{X}\|_F^2}$	0.8982	0.9982

Note: The image is recovered by computing $\widehat{\mathbf{x}} = \mathbf{I} \otimes \widehat{\mathbf{E}}\widehat{\mathbf{z}}$, where the subspace basis, $\widehat{\mathbf{E}}$, is learned from measurements (see (7)) and $\widehat{\mathbf{z}}$ is computed by solving $\widehat{\mathbf{z}} = \arg\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{M}(\mathbf{I} \otimes \widehat{\mathbf{E}})\mathbf{z} - \mathbf{y})\|^2$.

ratios were test. A common trend can be observed: the quality of reconstructed images (in terms of MPSNR) tends to decrease as the noise level increases. However, the proposed Hy-demosaicing method is relatively stable compared with other reconstruction methods.

Considering material identification via spectral analysis plays an important role in HSI applications, we show recovered spectral curves of representative materials in four images for evaluation. Fig. 9 shows measurements (marked as subsamples in the figure) of representative materials acquired by Hy-mosaicing sensing strategy and their spectra reconstructed by Hy-demosaicing. It is noted that subsamples are not exactly on the curves of noisy-free spectra (called references) because Gaussian noise (with standard deviation $\sigma = 0.02$) was added to subsamples. We can see from Fig. 9 that even with relatively low sampling ratio, 10%, the proposed Hy-demosaicing method is able to recover spectral vectors close to the noisy-free ones. The exploitation of spectral low-rankness of HSIs contributes to the success of spectral reconstruction.

C. Parameter Analysis of the Proposed Hy-Demosaicing

There are two critical parameters in the proposed Hy-demosaicing method: 1) the dimension of subspace, p and 2) the parameter of regularization, λ, which relates to the noise level. As we solve the subproblem (13) using

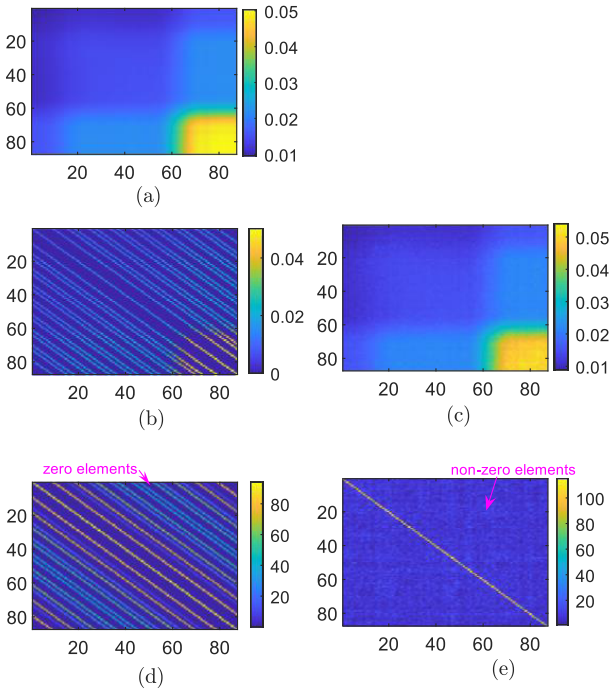


Fig. 13. True autocorrelation matrix (a) of spectral vectors in Pavia University data and its estimations (b) and (c) in two subsampling schemes.

FFDnet, the λ is corresponding to the standard deviation of noise, which was estimated by Foi's method[9] [47] in our experiments.

The optimal setting of the dimension of subspace, p, in the Hy-demosaicing method depends on the number of measurements, acquired by the subsampling. As we can see from Fig. 10, the optimal settings of p are 7, 3, 5, 5 for terrain data, Pavia University data, Washington DC Mall data, and Moffett field data, respectively. We remark that given the same sampling ratio and diverse number of bands, the number of measurements in four images are different. We acquired smallest amount of measurements for Pavia University data, thus its

[9]Foi's method to estimate noise standard deviation (AWGN model) from data of arbitrary dimensionality: www.cs.tut.fi/~foi/RiceOptVST/

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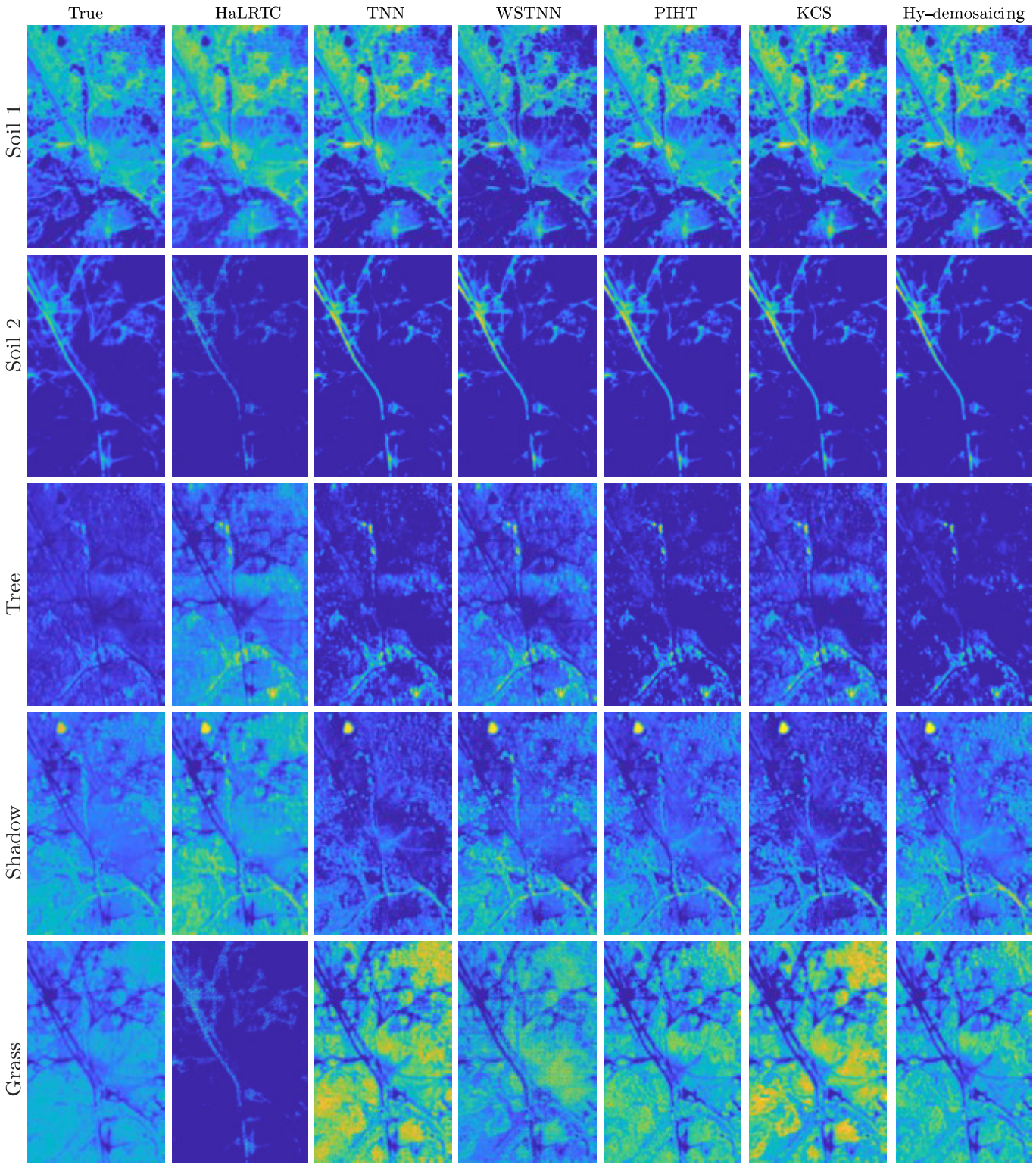


Fig. 14. Estimated abundance maps of terrain images reconstructed by different methods.

optimal p is much smaller than other images. According to the *law of large number*, sufficient number of measurements is required for a good estimate of subspace. When the dimension of subspace increases, more measurements are needed for subspace estimation. Thus given limited measurements, we suggest using a small value of p. Also, the underestimation of p (for example, $p = 2$) leads to low-quality reconstructed images since partial image information is lost through subspace projection. In our experiments, we simply set $p = 5$ for all images.

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(a) (b) (c)

Fig. 15. HSIs used in the experiments. (a) False color image of the HYDICE urban data set. (b) Subregion of urban data set used for anomaly detection. (c) Ground truth map of subregion of urban data set.

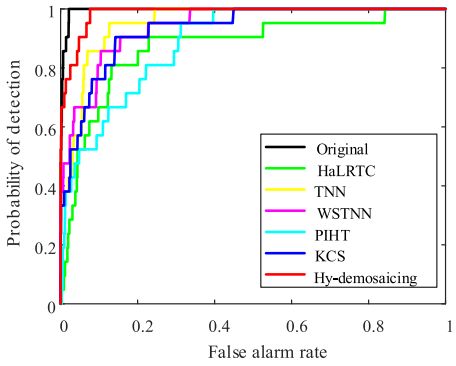


Fig. 16. ROC curves of CRD detector on original HYDICE urban image and images reconstructed by different methods.

D. Parameter Analysis of Structured SBSA

The SBSA can be implemented with different patterns, namely with different cell size K and different number of subsamples per pixel q. To satisfy the condition 2) in Claim (1), it requires larger settings of q and K. Regarding the effect of cell size, we take Pavia University data as an example and show MPSNRs of its reconstructed images as a function of cell size. We can see from Fig. 11(a) that MPSNR is practically constant when the cell size is greater or equal to 10×10. Given the same SR, the decrease in the size of the cell results in poor image reconstruction performance. To explain why this happens, we draw histograms of colors selected by our scheme in the Pavia University data. We can see from Fig. 12 that when the cell size is getting larger, the histogram of colors is closer to a uniform distribution. As we do not have prior knowledge that which color/spectral channels provide more important information than other colors, a uniform distribution of measured colors is preferred. However, when the cell size is too small, the uniform distribution of colors in the measurements is not guaranteed. The first histogram (with a cell size 5×5) in Fig. 12 is far away from a uniform distribution, implying some colors were under-sampled. This causes an unsatisfactory image reconstruction in Fig. 11(a). Considering the color balance in measurements, a setting of small cell size should be avoided. But we remark that a setting of very large cell size is not necessary in practice as larger cell size brings higher hardware complexity and does not improve image reconstruction accuracy significantly [see Fig. 11(a)].

The sensing strategy implemented with different settings of q leads to different sampling ratio. The increase in the number of subsamples per pixel, q, means more image information is acquired by the sensing strategy, which is beneficial to image reconstruction [see Fig. 11(b)]. But we remark that increasing number of subsamples per pixel causes larger measurement burden as well. Basically, the setting of q and K in the sensing strategy is a tradeoff between image reconstruction quality and measurement burden. A balanced setting of these sensing parameters should be discussed for specific applications.

E. Randomness of Subsampling Scheme

Instead of acquiring all samples in the spectral domain, we randomly select a few samples per pixel. One question now may be why we use a random scheme per pixel, not a predetermined scheme. For example, a predetermined scheme, "RGGB" pattern, is adopted in the Bayer CFA implemented in RGB cameras. An experiment using Pavia University data was conducted to show the impact of using different subsampling schemes. We compare our random scheme to a predetermined scheme. That is, for each pixel, we acquire colors equally distributed in the spectral domain. For example, indices of colors/bands selected in the first, second, third pixels are $[1, 10, 19, \ldots, 73]$, $[2, 11, 20, \ldots, 74]$, and $[3, 12, 21, \ldots, 75]$, respectively. As reported in Table V, using the same image reconstruction method, we were not able to recover the image from measurements acquired by the predetermined subsampling scheme. The main reason is that subsampling schemes affect the estimation of subspace basis. We measured the representation ability of subspaces by computing the relative power of clean pixels lying in the estimated subspace (denoted as $\gamma = \|\widehat{\mathbf{E}}\widehat{\mathbf{E}}^T\mathbf{X}\|_F^2/\|\mathbf{X}\|_F^2$). As shown in the final row of Table V, the representation ability of subspace learned from the predetermined scheme is worse than the subspace learned from the random scheme. To further explain why subsampling schemes affect the estimation of subspace, we visualize matrix $\sum_{k=1}^{K} \mathbf{a}_k\mathbf{a}_k^T$ in Fig. 13. In the predetermined scheme, the sparsity of $\sum_{k=1}^{K} \mathbf{a}_k\mathbf{a}_k^T$ [in Fig. 13(d)] results in an under-sampled autocorrelation matrix $\widehat{\mathbf{R}}$ [in Fig. 13(b)], which is a poor approximation of the true autocorrelation matrix [in Fig. 13(a)]. In the proposed scheme, the estimated autocorrelation matrix in Fig. 13(c) is close to the true one, implying that random subsampling scheme help image recovery.

F. Application in Hyperspectral Unmixing

We take hyperspectral unmixing as an example and conduct experiments using the semi-real terrain image to show the impact of Hy-mosaicing sensing scheme on hyperspectral unmixing. The SBSA was experimented on the semi-real terrain image with $q = 50$ and $K = 30$, yielding SR $= 30\%$. Gaussian i.i.d. noise with $\sigma = 0.04$ was added to the measurements.

Several image reconstruction methods were applied to recover the HSI from incomplete and noisy measurements. The

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TABLE VI
UNMIXING PERFORMANCE MEASUREMENTS ($NMSE_M$ AND $NMSE_A$) OF "VCA + FCLS" APPLIED TO THE RECONSTRUCTED TERRAIN IMAGES

Indexes	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-Demosaicing
$NMSE_M$	0.2427	0.0506	0.0781	0.0461	0.0504	**0.0440**
$NMSE_A$	0.5023	0.2506	0.1931	0.0967	0.2518	**0.0795**

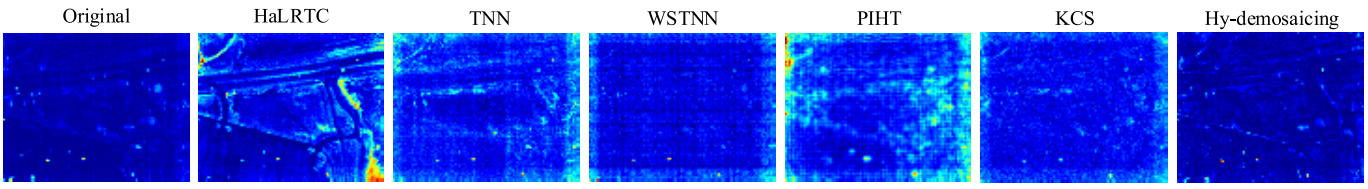


Fig. 17. Color anomaly maps obtained by CRD detector performing on different reconstruction images of HYDICE urban.

TABLE VII
AUC SCORES OF CRD DETECTOR ON DIFFERENT RECONSTRUCTION IMAGES OF HYDICE URBAN

Original	HaLRTC	TNN	WSTNN	PIHT	KCS	Hy-Demosaicing
0.9959	0.8717	0.9544	0.9413	0.8855	0.9298	0.9862

reconstruction performance is shown in Table I. We implemented spectral unmixing of the reconstructed images with vertex component analysis (VCA) estimating endmembers and fully constrained least squares (FCLS) estimating abundances. We remark that the endmembers found by VCA can be different in each run. To solve the non-reproducibility issue, we run VCA 30 times and output the endmembers, which yield largest simplex volume. Unmixing performance is measured by computing normalized mean square error (NMSE) of endmembers $\mathbf{M}$ and abundances $\mathbf{A}$, denoted as $NMSE_M$ and $NMSE_A$, respectively. We can see from Table VI that only the images reconstructed by PIHI and Hy-demosaicing methods yield low unmixing errors. Fig. 14 shows estimated abundance maps of reconstructed images. If we only focus on the abundance map of Grass, the one estimated from the reconstructed image of Hy-demosaicing is closest to the true one (see the first column of Fig. 14). This experiment shows that the image reconstructed from incomplete measurements is successfully used for spectral unmixing.

G. Application in HSI Anomaly Detection

The impact of the sensing strategy Hy-mosaicing and image reconstruction on HSI anomaly detection is evaluated in this section. The sensing strategy Hy-mosaicing was implemented on the well-known HYDICE urban data set with $K = 30 \times 30$ and $q = 16$ (leading to a sampling ratio SR = 10%). Several image reconstruction methods were applied to recover the HSI from the measurements.

A subregion of the urban image has been widely used for anomaly detection test [48]–[50]. The subregion and its ground truth map of anomalies are shown in Fig. 15(b) and (c). To evaluate whether the signal of anomalous pixels is lost in the reconstructed images, we performed anomaly detection using the well-known collaborative representation-based detector (CRD) [50] on the images reconstructed by different methods, namely, HaLRTC, TNN, WSTNN, PIHI, KCS, and Hy-demosaicing. The performance of anomaly detection is usually evaluated using a receiver operating characteristic (ROC) curve, which plots the detection rate against the false alarm rate. Fig. 16 illustrates the ROC curves for the application of CRD detector to the original urban image and reconstructed images. The area under per ROC curve (AUC) can be calculated. Table VII reports the corresponding AUC scores. The larger the AUC, the better detection performance. From Fig. 16 and Table VII, we can see that original urban image yields better results than reconstructed images. The degradation of anomaly detection performance happens because sampling ratio is only 10%. But if we only compare results of reconstructed images, the image reconstructed by Hy-demosaicing method yields the best detection result. The superiority of Hy-demosaicing also can be inferred from Fig. 17, showing the anomaly scores for all pixels yielding by CRD detector. The anomaly scores are normalized to within a range of 0–1. The anomaly map of the image obtained by Hy-demosaicing show a better separation between anomalies and background pixels.

V. CONCLUSION AND FUTURE RESEARCH LINES

This article introduces a joint design of a spectral subsampling strategy, termed Hy-mosaicing, and an image reconstruction algorithm, termed Hy-demosaicing. The sensing strategy is smart in the sense that it is hardware-friendly and the periodic structure of SBSA enable us to learn a spectral subspace from incomplete measurements, which is essential to the superiority of reconstruction algorithm. This sensing strategy is evaluated in experiments with one simulated HSI and four real HSIs, where the proposed Hy-demosaicing recovers data with high quality from highly incomplete measurements, taking into consideration the lightness and simplicity of the sensing strategy. The Hy-demosaicing is computationally much cheaper than other reconstruction methods, which is a desired feature in real-time applications. In the future, further work will be focused on the algorithm optimization of Hy-demosaicing so that it can be performed in real time.

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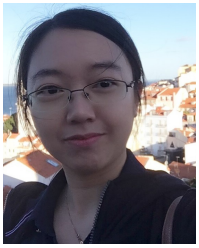
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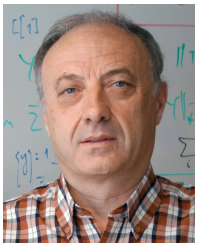
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